

AI Incident Observatory

1. Abstract

Nowadays, as artificial intelligence becomes increasingly integrated across industries, the number of AI related incidents and ethical concerns continues to grow. In this report, we present the AI Incident Observatory, a full-stack web application designed to collect AI related incidents from multiple public databases and provide tools for their visualization and analysis.

The platform also includes an ethics survey to help users find the AI model that best reflects their own ethical viewpoints. The Observatory is built using a lightweight Node.js and SQLite architecture with an interactive D3.js frontend, giving users a clear, data-driven view of global AI risks and trends.

2. Introduction

As AI is getting more popular, AI related incidents and ethical failures are increasing. Although several organizations track these incidents, the data remains scattered across many sources. At the same time, it is difficult for users to understand how closely AI models align with their own ethical views.

To address these challenges, we developed the AI Incident Observatory. The platform combines two main functions:

- Collecting and visualizing AI incidents from multiple public databases
- Providing a 100-question ethics quiz that matches users with the AI model whose responses most closely align with their own.

This report describes the system's architecture, data collection pipeline, deduplication process, and interactive visualization components.

3. Background and Related Work

Various organizations and groups have created databases to track AI failures.

- **AIID**: A community-driven catalog of incidents where AI systems have caused or nearly caused harm.
- **AIAAIC**: An independent public repository tracking AI controversy globally.
- **AVID**: A database focused specifically on vulnerabilities in ML models.
- **MITRE ATLAS**: A system that maps specific attacks against machine learning models.

While these separate databases are very useful, they all use different formats, interfaces, and designs. The AI Incident Observatory solves this by bringing all these sources into one central system.

4. System Methodology and Architecture

4.1 Planning and Architecture Design

The AI Incident Observatory adopts a lightweight, zero-build-step architecture prioritizing simplicity, fast execution, and ease of deployment.

- **Frontend:** HTML5, CSS3, and JavaScript. D3.js is used for data-driven DOM manipulation and physics simulations.
- **Backend:** A Node.js server that exposes RESTful JSON APIs to retrieve and sync data. Automated background tasks are scheduled using ``node-cron``.
- **Database:** SQLite3 in WAL mode offers highly efficient simultaneous reads and writes. A single integrated database file (``data/observatory.db``) makes deployment easy.

4.2 Data Search and Collection Strategy

A ``node-cron`` job runs every 6 hours, triggering a ``syncAll()`` orchestrator that contacts multiple sources:

- GraphQL APIs (for AIID)
- Public CSV exports (for AIAAIC)
- Parsed vulnerability repositories (for AVID and ATLAS)
- Curated keyword-based news feeds

4.3 Quality Assessment and Deduplication

Collecting data from multiple sources leads to duplicate reports. To keep the data clean, the system implements a two-pass deduplication algorithm:

- **Exact Match Verification:** Checks for existing records using ``external_id`` and ``source`` fields.
- **Fuzzy Title Matching:** Calculates the normalized Levenshtein distance on incident titles. If the difference score is lower than 0.15, the system marks the new record as a copy and drops it. This prevents the database from repeating data.

5. System Implementation

- **2D Canvas:** The main interface (*public/index.html*) uses a D3.js-powered physics simulation to display incidents as interactive nodes. Users can filter incidents by severity, company, category, geography, tags, source, and year, and explore them through five visualization modes: category, network, timeline, severity, and geography.
- **Analytics:** The analytics dashboard (*public/analytics.html*) transforms incident data into interactive charts, highlighting trends, incident categories, and affected countries.
- **AI Ethics Quiz:** The ethics module (*public/ethics.html*) contains 100 ethical questions and responses from 61 AI models. Users can complete the quiz to find the AI model that most closely matches their answers and compare their results with the overall AI results.
- **Database Schema and Migrations:** The SQLite database stores incident and ethics data in separate tables. Automatic migrations run at startup, allowing new data sources and schema updates to be added without manual intervention.

6. Limitations

6.1 Dependence on External Data Sources

We collect data from different sources. As a result, the system's ability to remain up to date depends on the availability of these external sources. If they change their system, our data collection may be broken.

6.2 Scalability Constraints

Now we use SQLite in WAL mode, which provides high performance for the platform's read-oriented workload. However, it has a limit for concurrent operations. If our site has more concurrent users, the database can be bottlenecked.

7. Conclusion

AI Incident Observatory is a platform to monitor AI risks. This platform includes an automatic data collection, an analytics module which visualizes data to show statistical patterns and trends, and a 100-question ethics quiz that measures the moral decisions of AI models.